

Riti Grover

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SUMMARY

Computer science undergraduate specializing in backend engineering and scalable AI systems. Delivered backend services in **Python**, **PostgreSQL** and **Firebase** for an offline-first AI edge platform at Fluxton, and applied machine learning through a **graph neural network** system that detects financial-crime patterns with a working forensic dashboard. Comfortable across **FastAPI**, **Flask**, **PyTorch**, **PyTorch Geometric**, **AWS (EC2, S3)**, **Docker**, and **Linux**. Strong foundation in data structures, algorithms, and object-oriented design, strengthened through internships, a research placement at IIT Roorkee, and self-directed projects.

EDUCATION

Manipal Institute of Technology

B.Tech. in Computer Science (Information Technology), CGPA: 8.70/10

Bengaluru, Karnataka

Jul. 2024 – May 2028

RS Memorial Higher Secondary School

Class XII (PCM): 81%

Katni, Madhya Pradesh

2023 – 2024

EXPERIENCE

AI / Backend Intern

Fluxton

Aug 2026 – Present

Bengaluru, Karnataka

- Delivering **6 REST API endpoints** in Python covering the full authentication flow (Firebase login, token refresh, logout, profile) for Fluxton's EV charging mobile app
- Designing a **3-table PostgreSQL schema** (users, sessions, rate-limiting counters) and adding **Firebase ID token verification** with short-lived JWT access tokens (15 min) and rotating, reuse-detected refresh tokens (7-day sessions)
- Shipping **7 commits** merged through 1 pull request in the project's first week, with 2 more feature branches in review for rate limiting and Firebase Functions deployment

Research Intern – AI-Driven Edge Computing for IoT

IIT Roorkee

May 2026

Roorkee, Uttarakhand

- Developed a real-time **resource allocation** approach for IoT edge systems using **Transformer-based** deep learning, working within real-world compute and latency constraints
- Implemented and benchmarked a hybrid **machine learning + optimization** framework against standard baselines, showing measurable gains across experiments

Creative Intern

Luscious Leopard

May 2025

Indore, Madhya Pradesh

- Redesigned **10+ website pages** and created **20+ branded digital assets**, turning business requirements into UI/UX improvements and content strategy that lifted user engagement by an estimated **15%** and strengthened brand consistency

PROJECTS

Smurfing Hunter | *PyTorch Geometric, Flask, D3.js, Graph Neural Networks*

[GitHub](#) | [Live Demo](#)

- Architected an end-to-end graph-based ML system for financial-crime detection, researching the methodology independently from data modeling through deployment
- Combined a 3-layer **Graph Convolutional Network (GCN)** in **PyTorch Geometric** with five heuristic pattern detectors (fan-out, fan-in, peeling chains, timing coordination, seed expansion) to flag suspicious wallet clusters across a 500-transaction synthetic Ethereum dataset
- Built a Flask REST API with 13 endpoints and a D3.js forensic dashboard, including automated Suspicious Activity Report (SAR) generation and wallet-level risk analytics, and deployed a live demo

Timetable AI | *FastAPI, Python, JavaScript, Constraint Satisfaction*

[GitHub](#) | [Live Demo](#)

- Designed a dependency-aware, constraint-driven scheduling architecture to resolve real-world timetable conflicts end-to-end
- Implemented **3 backtracking CSP configurations** (plain, with MRV heuristics and with MRV plus forward checking) and added an in-app performance table comparing execution time, recursive calls and constraint checks live on every solve
- Built a modular **FastAPI** backend and a responsive JavaScript frontend, tested up to **6 subjects, 4 time slots and a pool of 20 students** through a sample generator

COURSES AND CERTIFICATIONS

Data Analytics Foundations (<i>DeepLearning.AI</i>)	<u>certificate</u>
Fundamentals of Java Programming (<i>Board Infinity</i>)	<u>certificate</u>
Integrating with Structured Databases (<i>IBM SkillsBuild</i>)	<u>certificate</u>
Linux on IBM LinuxONE (<i>IBM SkillsBuild</i>)	<u>certificate</u>

SKILLS

Programming: Python, Java, C
Backend: FastAPI, Flask, REST APIs, Firebase Authentication, JWT, API Development, Backend Development
Databases: SQL, PostgreSQL, Database Design, Schema Design
ML/AI: Machine Learning, Deep Learning, Transformers, Graph Neural Networks (GNN), PyTorch, PyTorch Geometric
Cloud/DevOps: AWS (EC2, S3), Git, Linux, Docker
CS Fundamentals: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Operating Systems (OS), DBMS
Data/Visualization: Data Analysis, Data Visualization, Excel, D3.js
Languages: English, Hindi

CLUBS, COMMITTEES AND CO-CURRICULAR ACTIVITIES

Vice President, LitSoc – Literature Society | **Social Media Manager**, OtakuSpot – Pop Culture Club | **Executive Member**, IEEE Computer Society | **Organising Committee**, TechSolstice – Technical Fest | **Content Writer**, PCB – Photography Club | **Marketing Team**, Radar – Robotics Club
MIT Bengaluru – 2024 to 2026